

CIS 658 Web Architectures

Cloud DataBase

Firestore User Authentication



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Firebase: A collection of many products

- Authentication
- Cloud Firestore
- Cloud Storage
- Cloud Functions
- ...

Setup: Install and Initialize Firebase

```
yarn init -y  
yarn add firebase
```

OR

```
npm init -y  
npm install firebase
```

```
import { createApp } from "vue";  
import App from "./App.vue";  
import router from "./router";  
import { initializeApp } from "firebase/app";  
  
const firebaseConfig = {  
  apiKey: "your-api-key-goes-here",  
  authDomain: "your-project-name.firebaseio.com",  
  projectId: "your-project-id",  
  storageBucket: "your-project-name.appspot.com",  
  messagingSenderId: "xxxxxxxxxxxx",  
};  
  
initializeApp(firebaseConfig);  
createApp(App).use(router).mount("#app");
```

main.ts

Setup: Use Auth Functions

```
yarn init -y  
yarn add firebase
```

OR

```
npm init -y  
npm install firebase
```

```
import {  
  getAuth,  
  signInWithPopup,  
  createUserWithEmailAndPassword,  
  signInWithEmailAndPassword,  
  sendEmailVerification,  
  signOut,  
} from "firebase/auth";  
  
const auth = getAuth();  
createUserWithEmailAndPassword(auth, "user_email", "user_password");  
sendEmailVerification(auth.currentUser);  
signInWithEmailAndPassword(auth, "user_email", "user_password");  
signInWithPopup(auth, _____);  
signOut(auth);
```

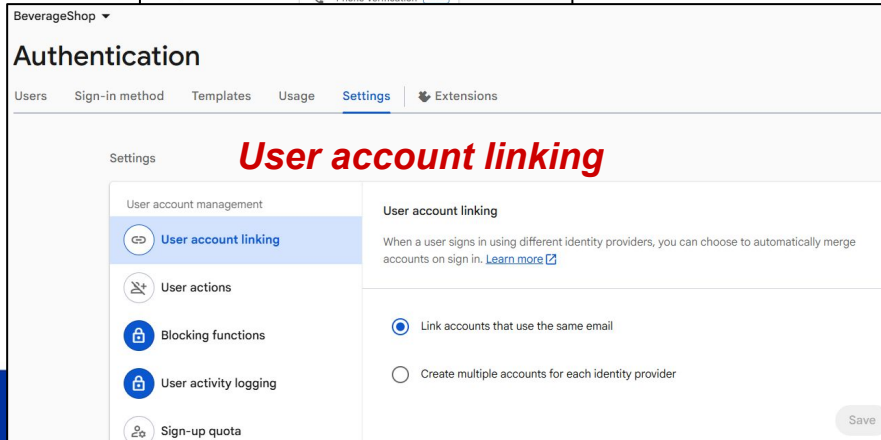
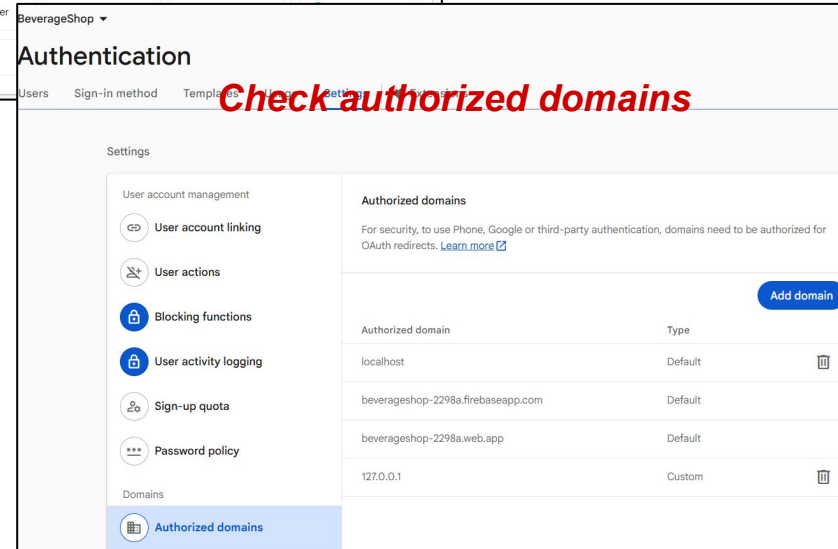
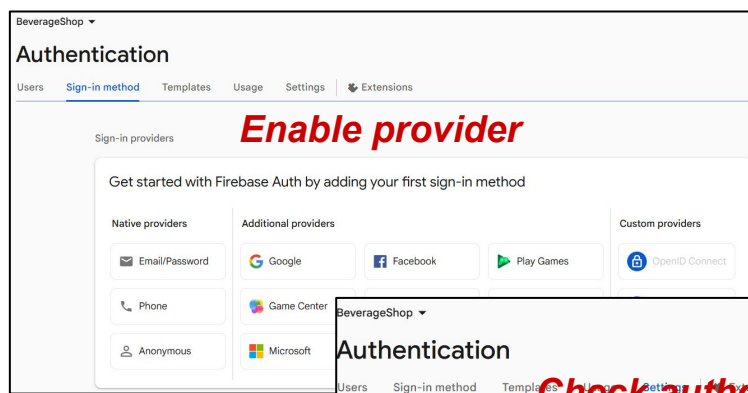
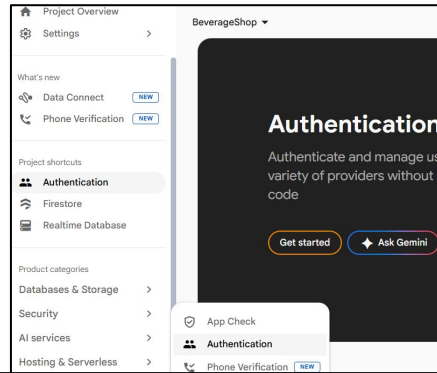
login.vue

Authentication Options

- Email/Password
- GitHub accounts
- Google accounts
- ...

- **Firebase Authentication Web docs.**

What to configure in Firebase Console



Authentication Functions are async
Use Promise.then or await to handle the result

Firebase Auth: Create A New Account

```
import {
  getAuth,
  createUserWithEmailAndPassword,
  UserCredential,
} from "firebase/auth";
const auth = getAuth();

createUserWithEmailAndPassword(auth, "me@sample.com", "1q2w3e4r5")
  .then((cred: UserCredential) => {
    sendEmailVerification(cred.user);
    console.log("Verification email has been sent to", cred.user?.email);
    auth.signOut();
  })
  .catch((err: any) => {
    console.error("Oops", err);
  });
```



What will happen after calling the
`createUserWithEmailAndPassword` function?

Firebase Auth: Signin With Email

```
import {
  getAuth,
  signInWithEmailAndPassword,
  UserCredential,
} from "firebase/auth";
const auth = getAuth();

signInWithUserWithEmailAndPassword(auth, "me@sample.com", "1q2w3e4r5")
  .then((cred: UserCredential) => {
    if (cred.user?.emailVerified) console.log("Signed in as", cred.user?.email);
    else {
      console.log("Please verify your email first");
      auth.signOut();
    }
  })
  .catch((err: any) => {
    console.error("Oops", err);
  });
```

Firestore Auth: Sign In With Providers

```
import { getAuth, GoogleAuthProvider, signInWithPopup } from "firebase/auth";
const auth = getAuth();

const provider = new GoogleAuthProvider();

signInWithPopup(auth, provider)
  .then((result) => {
    const cred = GoogleAuthProvider.credentialFromResult(result);
    console.log("Signed in as", cred.user?.email);
  })
  .catch((err: any) => {
    console.error("Oops", err);
  });
```

Make sure you have added the login provider in the Authentication Dashboard

Monitor Authentication in Background

```
import { getAuth, User, onAuthStateChanged } from "firebase/auth";
const auth = getAuth();

onAuthStateChanged(auth, (user: User | null) => {
  if ((currentUser == null) & (user != null)) {
    currentUser = user;
    /* User just logged in */
  } else if ((currentUser != null) & (user == null)) {
    /* User just logged out, do clean up work */
  }
});
```

a list of available properties

Use onAuthStateChanged if you need to monitor login status in background

3rd party Account Providers

Account Provider	Provider Class
Facebook	FacebookAuthProvider
GitHub	GithubAuthProvider
Google	GoogleAuthProvider
Phone	PhoneAuthProvider
Twitter	TwitterAuthProvider

Example: GitHub SignIn

Step A: Enable Github Login

BeverageShop ▾

Authentication

Users Sign-in method Templates Usage Settings Extensions

Sign-in providers

 GitHub

☒ Enable

Client ID

ⓘ A client ID is required

Client secret

ⓘ A client secret is required

To complete set up, add this authorization callback URL to your GitHub app configuration.
[Learn more](#)

https://beverageshop-2298a.firebaseio.com/_/auth/handler

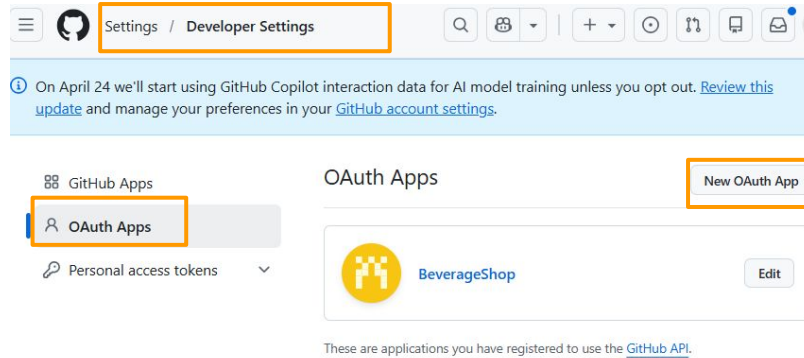
Copy callback URL

Cancel Save

Do this step from Firebase Authentication Dashboard

Step B: Create an OAuth App (On Github Settings)

The following page is under Settings => Developer Setting



*Create a new
Github OAuth App*

Do this step from GitHub Developer Settings

Step D: Copy Client ID & Client Secret

Register a new OAuth application

Application name *

Something users will recognize and trust.

Homepage URL *

The full URL to your application homepage.

Application description

Application description is optional

This is displayed to all users of your application.

Authorization callback URL *

Your application's callback URL. Read our [OAuth documentation](#) for more information.

☐ Enable Device Flow

Allow this OAuth App to authorize users via the Device Flow.
Read the [Device Flow documentation](#) for more information.

Register application Cancel

app name to show to the user at log in time

enter a valid URL and this can be changed later

*copy and paste from
Firebase Authentication
(from Step A)*

Settings / Developer settings / BeverageShop

General

Optional features

Advanced

BeverageShop

 Yong-Zhuang owns this application. [Transfer ownership](#)

You can list your application in the [GitHub Marketplace](#) so that other users can discover it. [List this application in the Marketplace](#)

2 users [Revoke all user tokens](#)

Client ID

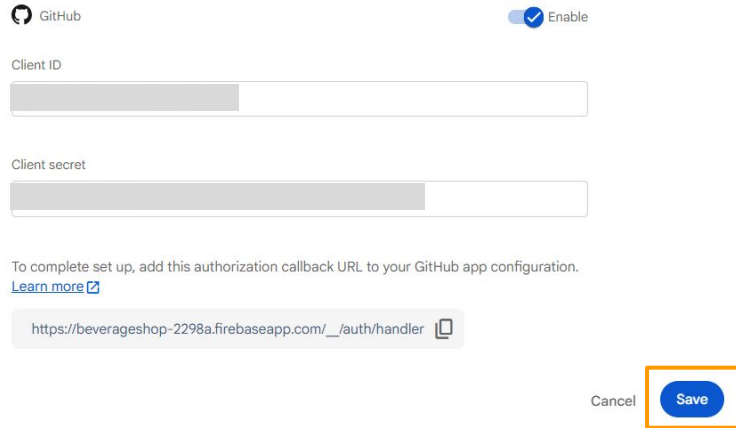
Client secrets [Generate a new client secret](#)

 Client secret

Last used within the last 5 months

You cannot delete the only client secret. Generate a new client secret first. [Delete](#)

Step D: Copy Client ID & Client Secret



The screenshot shows the GitHub OAuth app configuration page. At the top, there is a GitHub logo and an 'Enable' toggle switch which is turned on. Below this, there are two input fields: 'Client ID' and 'Client secret'. Both fields have a grey bar at the beginning, indicating that content has been copied. Below the input fields, there is a text instruction: 'To complete set up, add this authorization callback URL to your GitHub app configuration.' followed by a link 'Learn more'. Below this instruction, there is a text box containing the URL 'https://beverageshop-2298a.firebaseio.com/_/auth/handler' and a copy icon. At the bottom right, there are two buttons: 'Cancel' and 'Save'. The 'Save' button is highlighted with an orange border.

GitHub ☒ Enable

Client ID

Client secret

To complete set up, add this authorization callback URL to your GitHub app configuration.
[Learn more](#)

`https://beverageshop-2298a.firebaseio.com/_/auth/handler` 

Cancel **Save**

Do this step from Firebase Authentication Dashboard

Exercises: Try User Authentication

1. Open Your Firebase Project

- Go to Firebase Console and open your **BeverageShop** project
- In the left sidebar, go to **Build** → **Authentication**
- Click **Get started**

2. Enable a Sign-In Method

- Open the **Sign-in method** tab
- Enable **Email/Password**
- Save the changes

3. Check Authorized Domains

- Open the **Settings** tab in Authentication
- Find **Authorized domains**
- Make sure your local development domain is listed, such as **localhost**, **127.0.0.1**